

AKSHAY ASHOK

M.Sc. Data Science & Artificial Intelligence | Saarland University

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PROFILE

Master's student in Data Science & AI (5th semester) who ships measured, reproducible ML: a first-author Springer paper in computer vision, a reproducibility study of a multimodal LLM, an Arabic OCR and translation benchmark, a deployed NLP scoring service and a process-mining study of a 1.6M-event purchase-to-pay log. Strongest in evaluation design and failure analysis: finding where a reported result does less work than it appears to. Open to a Master's thesis topic for summer 2027 and research collaborations.

EXPERIENCE

Working Student, Data Science & Process Intelligence · ORBIS Value+ · Saarbrücken Oct 2026 – Present

- Data science in the Process Intelligence team, part-time alongside the M.Sc.

Student Consultant (AI / Data Science) · Lingomatch GmbH · Saarbrücken Apr – Jun 2026

- Built and benchmarked the ML backend of a document-triage feasibility study for a Gov/Justice-Tech platform (UdS Innovation Center): photos of Arabic documents to German gist translations on an 8 GB laptop GPU.
- Evaluated Surya OCR on KITAB-Bench (per-sample CER/WER, confidence calibration); tested PaddleOCR as an independent second engine for disagreement-based flagging and showed it disagrees with Surya even on perfect reads.
- Benchmarked Opus-MT vs. NLLB vs. TranslateGemma; caught the CC-BY-NC licence ruling the best baseline out of production and recommended an Apache-2.0 model with comparable BLEU on short sentences at 1/9 the memory.
- Designed confidence-based failure routing that flagged 4/4 catastrophic OCR failures; presented the live demo to the CEO.

github.com/Axwolf13/arabic-doc-triage-research · write-up: axwolf13.github.io/writing/arabic-ocr-benchmarks

Research Intern (Computer Vision) · National Institute of Technology, Puducherry Jan – Apr 2024

- Built a real-time yoga pose estimation pipeline with a hybrid MoveNet Thunder (training) + Lightning (inference) architecture: 11 asanas at 98.8% test accuracy (98.4% on the shipped quantized model) at 30+ FPS; curated the 3,435-image dataset.
- First-authored a peer-reviewed publication in *Sādhana* (Indian Academy of Sciences, Springer, 2025); released the code, the trained model and a live in-browser demo (TensorFlow.js) with a verification script that reproduces the 98.4%.

doi.org/10.1007/s12046-025-02788-w · live demo: axwolf13.github.io/demo · code: github.com/Axwolf13/pose-estimation-movenet

Web Development Intern · Zoho Corporation, Chennai, India Aug – Sep 2023

- Built a full-stack library management web app with a content-based book recommender (HTML/CSS, JavaScript, SQL).

SELECTED PROJECTS

Purchase-to-pay process mining · public BPI Challenge 2019 log

- Analysed 251,734 purchase-order items (1.6M events, 42 activities) in pandas: throughput, matching-rule compliance, rework, automation and process variants, with tested rule checks and CI.
- Found 96% of invoice handling is manual and 22% of items need a payment block lifted, half of them in one spend area; flagged 654 items whose invoice was cleared before any goods receipt.

github.com/Axwolf13/p2p-process-mining · write-up: axwolf13.github.io/writing/purchase-to-pay

MolCA reproduction: which input does a molecule LLM believe? · graduate seminar, Prof. Klakow

- Reproduced MolCA (EMNLP 2023) on two machines: 62.32 and 62.77 BLEU-2 against the paper's 62.0, 93.8% of captions character-identical, all 32 published retrieval metrics within 0.5 points; 14 patches to the code, several for unreported bugs.
- Designed the modality-conflict test the paper never ran: given one molecule's SMILES and another's graph, the model follows the graph in about 90% of decisive cases; a wrong graph costs 2.5× what wrong text does (paired bootstrap, every interval excludes zero); controlled for 49% train-test contamination on a second dataset.

github.com/Axwolf13/molca-reproduction · write-up: axwolf13.github.io/writing/molca

DODI: deception scoring for Terms-of-Service documents, shipped as a live web service

- Deterministic index of how hard a ToS hides that "Buy now" means "revocable licence"; temporal study of 10 platforms across 2015–2024, compared against ToS/DR expert grades.
- Cross-checked with three LLM judges from two vendors: they barely agree with the index ($p = 0.05$ to 0.26) and expose a false positive on honest subscriptions, which v2 removes by weighing each contract against what the store promised.
- Dockerized FastAPI service with tests and CI (dodi-web.onrender.com) plus an MCP server tested in Copilot Studio.

EDUCATION

M.Sc. Data Science & Artificial Intelligence · Saarland University Oct 2024 – Present

Coursework: Statistics with R, Human-Computer Interaction, Data and Society. Seminar: ML for Language Processing (MolCA).

B.Tech. Artificial Intelligence & Data Science · Bannari Amman Institute of Technology Oct 2020 – Apr 2024

CGPA: 8.65/10. Thesis: LipNet-based visual speech recognition as a personal communication aid for hearing-impaired users.

TECHNICAL SKILLS & LANGUAGES

Languages & tools: Python, R, SQL, JavaScript; PyTorch, TensorFlow/Keras, Hugging Face Transformers, scikit-learn, spaCy, OpenCV, pandas, NumPy; FastAPI, Docker, CI/CD (GitHub Actions), Git, HTCondor, LaTeX.

LLM work: OpenAI, Anthropic & Gemini APIs, prompt design, structured outputs, LLM-as-judge evaluation, MCP server development, embedding-based retrieval, agentic workflows (Claude Code, Cursor).

Methods: model benchmarking and evaluation design (BLEU, CER/WER, calibration, paired bootstrap CIs), reproducibility and ablation studies, classical ML & deep learning, mixed-effects models, event-study design, NLP and OCR pipelines.

Spoken languages: English (C1), German (A2, targeting B1), Tamil (native), Hindi (conversational).